Phase-Modulated Compound Sparsity: Strategy-Dependent Severity Gradients in Reasoning Models

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Abstract

Compound sparsity—simultaneously pruning attention heads and evicting KV cache entries—is essential for efficient LLM inference but produces super-additive quality degradation whose structure is poorly understood. Prior work treats compound interaction magnitude as approximately constant across the generation process. We show that this assumption is wrong: compound sparsity severity is an emergent phase-strategy interaction that varies by over an order of magnitude across reasoning phases.

We evaluate compound sparsity on DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct (both Qwen2 architecture, 7B parameters) using 50 MATH-500 problems, measuring the Interaction Index—a super-additivity metric—across quantile-segmented reasoning phases at sparsity levels $s \in \{0.15, 0.20, 0.25, 0.30\}$. Under importance-based KV cache eviction, the median Interaction Index difference between low-sparsity and high-sparsity reasoning phases increases monotonically from 0.39 at s=0.15 to 14.59 at s=0.30, with paired Cohen's d_z growing from 0.44 (small) to 0.80 (large); Bonferroni-corrected permutation tests yield $p < 0.005$ at all four sparsity levels. Random eviction produces no significant phase modulation in 7/8 conditions; the single exception (Qwen s=0.30, p=0.012) shows negligible effect size (d_z=−0.04), confirming that the severity gradient is strategy-specific, not an inherent property of compound sparsity. At s=0.25, a crossover emerges: importance-based eviction outperforms random in high-sparsity phases ($\widetilde{\mathrm{II}} = 0.032$ vs. 0.818) yet underperforms in low-sparsity phases ($\widetilde{\mathrm{II}} = 7.196$ vs. 1.363)—a qualitative reversal in strategy ranking. The pattern replicates on Qwen2.5-7B-Instruct at $s \geq 0.20$ ($d_z \geq 0.55$; Spearman $\rho = 1.0$ across sparsity levels; gradient permutation $p = 0.043$).

We propose entropy-discrimination collapse as an explanatory mechanism: head pruning flattens cumulative importance scores in phases where attention is already broadly distributed, degrading the discriminative power of importance-based eviction in a phase-dependent manner. These findings challenge the phase-invariance assumption in compound sparsity deployment and motivate phase-aware compression budgets for reasoning models.

1 Introduction

Long-context language models place acute pressure on the key–value (KV) cache, which grows linearly with sequence length and dominates inference memory at scale (Kwon et al., 2023; Pope et al., 2023). Two complementary responses have emerged: *attention head pruning*, which removes structurally redundant heads (Voita et al., 2019; Michel et al., 2019), and *KV cache eviction*, which discards low-importance token positions from the cache (Li et al., 2024; Feng et al., 2025). Deploying both simultaneously—*compound sparsity*—is attractive because the compression gains multiply, but the resulting quality degradation can be super-additive: the joint loss exceeds the sum of individual losses (Liao et al., 2026; Mozaffari et al., 2025).

Prior work has characterized compound interaction as a property of the sparsity level—higher sparsity yields more interaction—but treats the interaction magnitude as roughly constant across the generation process (Liao et al., 2026; von Rad et al., 2026). This assumption is problematic for reasoning-capable models, whose generations are structurally heterogeneous: a single chain-of-thought trace alternates between phases of broadly distributed attention (exploratory reasoning) and phases of sharply concentrated attention (structured computation) (Ramachandran et al., 2026; Shi et al., 2026). If compound interaction severity varies across these phases, then a uniform sparsity budget—the standard deployment practice—may

be safe in one phase yet destructive in another.

We investigate whether compound sparsity severity is modulated by reasoning phase and eviction strategy. Our central finding is that it is, and strongly so: under importance-based KV cache eviction, the Interaction Index—a measure of super-additive compound degradation—is concentrated in the low-sparsity phase (where attention distributes broadly) and largely absent in the high-sparsity phase (where attention is focused). This *severity gradient* grows monotonically with the sparsity budget: on DeepSeek-R1-Distill-Qwen-7B, the median phase difference in Interaction Index rises from 0.39 at s=0.15 to 14.59 at s=0.30, with paired effect sizes increasing from d_z=0.44 (small) to d_z=0.80 (large); Bonferroni-corrected permutation tests confirm significance at all four sparsity levels on this model ($p < 0.005$). The pattern replicates at $s \geq 0.20$ on Qwen2.5-7B-Instruct ($d_z \geq 0.55$) with attenuated absolute magnitude, though the effect at s=0.15 is marginal (d_z=0.28, bootstrap CI includes zero). Critically, random eviction produces no such gradient: 7/8 conditions across both models fail to exhibit a significant phase difference (the single exception—Qwen s=0.30—has negligible effect size, d_z=−0.04), confirming that the severity gradient is an emergent property of the interaction between importance scoring and phase-dependent attention structure, not an artifact of compound sparsity per se.

We propose *entropy-discrimination collapse* as an explanatory mechanism. Head pruning forces surviving heads to redistribute attention mass, increasing attention entropy. In low-sparsity phases, where attention is already broadly distributed, this redistribution flattens the cumulative importance scores used for KV eviction, degrading their discriminative power. Importance-based eviction then degenerates into a biased pseudo-random process that can inadvertently remove critical positions—amplifying compound severity. In high-sparsity phases, attention concentration is preserved after pruning, importance scores remain discriminative, and eviction remains well-targeted. Random eviction is immune to this mechanism because its decisions are independent of attention patterns.

Our contributions are as follows:

1. We identify a **phase–strategy interaction** in compound sparsity: importance-based eviction exhibits a severity gradient across reasoning phases that is absent under random eviction (§5).
2. We show that this severity gradient **increases monotonically with sparsity** and replicate the finding at moderate-to-high sparsity ($s \geq 0.20$) across two models with different training regimes (distillation vs. instruction tuning), establishing robustness within the Qwen2 architecture family (§5).
3. We propose **entropy-discrimination collapse** as a mechanistic explanation, linking head pruning to the degradation of importance-score discriminability in a phase-dependent manner (§6).

These findings challenge the implicit assumption that compound interaction magnitude is phase-invariant, and suggest that sparsity budgets should be allocated in a phase-aware manner—tightening compression during low-sparsity reasoning phases where compound interaction is most severe. The remainder of this paper is organized as follows: §2 surveys related work on compound compression, KV cache eviction, and phase-aware inference; §3 formalizes compound sparsity and introduces the Interaction Index; §4 describes our experimental setup; §5 presents the main results; §6 analyzes the entropy-discrimination collapse mechanism; and §7 discusses implications and limitations.

2 Related Work

2.1 Compound and Joint Compression

Combining multiple compression axes produces non-additive errors. (Liao et al., 2026) report super-additive degradation when combining channel pruning with token eviction, with up to 65% quality loss exceeding individual component effects—the first explicit measurement of this phenomenon in KV cache sparsity. (Mozaffari et al., 2025) formalize compound error for quantization and sparsity, deriving an invertible saliency function for low-rank compensation and confirming that aggregated error is strictly non-additive. (von Rad et al., 2026) evaluate compound compression across 40+ benchmarks, revealing that reasoning and multilingual capabilities degrade disproportionately relative to factual recall. The order of compression stages is itself consequential: (Chhawri et al., 2025) show that early quantization causes irreversible information loss, making P→KD→Q strictly preferable. At a more fundamental level, (Ma et al., 2026) characterize independent phase transition points per compression dimension, implying that the joint safe region is the intersection—not the union—of per-dimension safe zones. On the structural

side, GRASPrune (Wang et al., 2026) learns global budget-constrained gates to jointly prune FFN channels and attention heads, but treats the pruning decision as static across generation. These works establish that compound interaction produces non-additive errors, but share an implicit assumption: the interaction magnitude is either constant across the generation process or varies only by task—not by reasoning phase or eviction strategy. How compound severity shifts across reasoning phases within a single generation remains unexamined.

2.2 KV Cache Eviction

Most eviction methods rely on importance scoring, yet accumulating evidence questions its reliability. Ada-KV (Feng et al., 2025) introduce per-head adaptive budget allocation, outperforming uniform distribution but assuming stable importance across generation. SnapKV (Li et al., 2024) identifies important KV positions per head from an observation window at the end of the prompt, compressing the cache before generation begins—effective for long contexts but committing to a fixed importance snapshot that cannot adapt across reasoning phases. (Feng et al., 2026) show that standard mean-based importance aggregation is fragile, with worst-case quality loss 2.3–4.3$\times$ higher than expected; their defensive aggregation mitigates but does not eliminate the instability. (Chen et al., 2024) combine proxy-token eviction with random eviction to counteract attention bias, improving robustness over pure importance-based methods; yet the eviction policy remains uniform throughout generation. (Gu et al., 2025) derive a second-order joint K-V saliency score from optimal brain damage theory, establishing that key pruning impacts quality more than value pruning—a useful asymmetry, though the score remains time-invariant. (Joo et al., 2025) achieve 70% unstructured KV sparsity through per-token magnitude pruning without quality loss, demonstrating that fine-grained within-entry sparsity and coarse token eviction are complementary axes. RetentiveKV (Liu et al., 2026) identifies *deferred importance*: tokens with low initial salience can become pivotal later, undermining single-snapshot importance estimates. (Sood et al., 2026) further show that any single importance metric exhibits greedy bias, systematically missing structurally relevant tokens. A recurring theme is that importance estimates become unreliable under specific conditions, but these conditions are characterized in terms of aggregation functions or metric choice—not the reasoning phase during which eviction occurs, nor the interaction with concurrent head pruning.

2.3 Phase-Aware Inference

Several concurrent works introduce phase-awareness into LLM inference. ThinKV (Ramachandran et al., 2026) partitions generation into phases and applies different KV cache strategies per phase, achieving strong single-axis results that motivate our phase decomposition—but without compound interaction. (Chitty-Venkata et al., 2026) manage block-wise eviction with separate prefill and decode policies, a coarse two-phase scheme that does not extend to within-generation reasoning phases. (Shi et al., 2026) discover that layer pruning sensitivity depends on whether a layer operates in the "silent" or "decisive" phase of representation formation, with silent-phase pruning causing collapse—direct evidence that compression tolerance is phase-modulated. (Monjur et al., 2026) provide sign reversal evidence: unstructured pruning combined with test-time scaling can exceed unpruned baselines, suggesting that sparsity interacts constructively with extended reasoning under certain conditions. (Hu et al., 2025) report non-monotonic token importance across decoding steps, and (Nawrot et al., 2025) find no universally optimal sparse attention strategy across tasks and generation stages.

These works establish that compression tolerance is phase-dependent, but each examines either phase-aware compression along a single axis or compound compression without phase conditioning. Our work bridges this gap by quantifying how compound sparsity severity varies across reasoning phases and how this variation is modulated by eviction strategy—specifically, the phase-dependent severity gradient that distinguishes importance-based eviction from random baselines in compound settings.

3 Background and Problem Setup

We formalize compound sparsity as the simultaneous application of attention head pruning and KV cache eviction, define the Interaction Index to quantify their non-additive coupling, and introduce phase segmentation to localize the analysis across reasoning stages.

3.1 Compound Sparsity Formulation

Consider a Transformer language model with L layers, each containing H attention heads. Let $s \in (0, 1)$ denote a unified sparsity budget applied to both compression axes.

Attention head pruning. For each layer ℓ, heads are ranked by Frobenius norm importance $w_j^{(\ell)} = \|\mathbf{A}_j^{(\ell)}\|_F$, where $\mathbf{A}_j^{(\ell)} \in \mathbb{R}^{T\times T}$ is the attention matrix for head j (Voita et al., 2019; Michel et al., 2019). A binary mask $m_j^{(\ell)} \in \{0, 1\}$ retains the top-$(1-s) \cdot H$ heads, zeroing the output of pruned heads:

$$\tilde{\mathbf{o}}_j^{(\ell)} = m_j^{(\ell)} \cdot \mathbf{o}_j^{(\ell)}. \tag{1}$$

KV cache eviction. For a sequence of length T, a position mask $\mathbf{m}_k \in \{0, 1\}^T$ retains the top-$(1-s) \cdot T$ positions, with the BOS token always preserved. Evicted positions are blocked via attention masking:

$$\tilde{a}_{ij} = a_{ij} \cdot m_{k,j}, \quad \forall\, i, j. \tag{2}$$

We compare two strategies for constructing $\mathbf{m}_k$:

- **Importance-based:** Each position j is scored by its cumulative attention mass with position normalization:

$$\mathrm{imp}(j) = \frac{\sum_{\ell,h,i} a_{ij}^{(\ell,h)}}{T - j}, \tag{3}$$

  where the denominator compensates for causal bias—earlier positions are attended to by more queries (Ramachandran et al., 2026; Feng et al., 2026).
- **Random:** Positions (excluding BOS) are evicted uniformly at random.

Compound sparsity. Applying both mechanisms simultaneously at sparsity level s (pruning fraction $p = s$, eviction fraction $q = s$) yields *compound sparsity*. The two axes are not independent: head pruning redistributes attention mass among surviving heads, which alters the importance ranking used for KV cache eviction. Prior analyses of compound compression (Liao et al., 2026) report that the joint effect is generally super-additive but treat the interaction magnitude as a fixed property of s. We investigate whether the eviction strategy and reasoning phase introduce additional structure.

3.2 Interaction Index

To quantify the interaction between head pruning and KV cache eviction, we define the *Interaction Index* (II) following the factorial interaction framework from experimental design.

For a given text segment, let PPL_d, PPL_a, PPL_k, and PPL_b denote the perplexity under the dense (no sparsity), attention-pruning-only, KV-eviction-only, and compound conditions, respectively. Perplexity is computed as $\mathrm{PPL} = \exp\big(\frac{1}{|S|}\sum_{t\in S}\mathcal{L}_t\big)$, where $\mathcal{L}_t$ is the per-token cross-entropy loss over evaluated positions S. Normalizing by the dense baseline yields relative degradation ratios $r_c = \mathrm{PPL}_c \,/\, \mathrm{PPL}_d$ for $c \in \{a, k, b\}$, and the Interaction Index:

$$\mathrm{II} = r_b - r_a - r_k + 1. \tag{4}$$

Under additive independence, $r_b \approx r_a + r_k - 1$, yielding $\mathrm{II} \approx 0$. The sign of II determines the interaction regime:

- $\mathrm{II} > 0$: **super-additive**—compound degradation exceeds the sum of individual effects.
- $\mathrm{II} < 0$: **sub-additive**—the compound effect is less than the sum of parts.
- $\mathrm{II} = 0$: **additive**—the two mechanisms do not interact.

Choice of additive null. Equation 4 posits an additive null $r_b = r_a + r_k - 1$ rather than multiplicative independence $r_b = r_a \cdot r_k$. We adopt the additive form because head pruning and KV cache eviction operate on disjoint model dimensions—structural (which heads to retain) versus positional (which tokens to cache)—so their degradations sum rather than multiply at the sparsity levels we study ($s \leq 0.30$). We verify in Appendix A that qualitative conclusions are unchanged under the multiplicative null.

Normalizing by PPL_d ensures cross-problem comparability. Because the II distribution is heavily right-skewed, we report medians as the central tendency.

Severity gradient. To capture how II varies across reasoning phases (defined in §3.3), we define the *severity gradient*:

$$G = \frac{\widetilde{\mathrm{II}}_{low}}{\widetilde{\mathrm{II}}_{high}}, \tag{5}$$

where $\widetilde{\mathrm{II}}_{low}$ and $\widetilde{\mathrm{II}}_{high}$ are the median interaction indices in the low-sparsity and high-sparsity phases, respectively. $G \gg 1$ indicates that compound interaction concentrates in low-sparsity phases; $G \approx 1$ indicates phase-invariant degradation. When $\widetilde{\mathrm{II}}_{high} < \varepsilon$ for a small threshold $\varepsilon > 0$, we report $G = \infty$.

3.3 Phase Segmentation

Reasoning-capable LLMs produce token sequences with heterogeneous attention structure: some positions exhibit highly concentrated attention over a few key tokens, while others distribute attention broadly across the context. We partition each generated sequence into three phases based on the local *attention sparsity profile*, adapting the phase-aware methodology of ThinKV (Ramachandran et al., 2026).

Attention sparsity. For each query position i and head h, we compute the fraction of key positions receiving negligible attention:

$$\sigma_i^{(h)} = \frac{1}{T} \sum_{j=1}^{T} \mathbf{1}\left[a_{ij}^{(h)} < 0.01 \cdot \max_k a_{ik}^{(h)} \right]. \quad (6)$$

The per-position sparsity $\sigma_i = \frac{1}{H} \sum_h \sigma_i^{(h)}$ averages across all heads.

Quantile-based segmentation. Tokens are partitioned into three groups by the tercile boundaries of $\{\sigma_i\}_{i=1}^{T}$:

- **High-sparsity phase** (top tercile of σ): attention concentrates on few tokens; heads are largely redundant, making both pruning and eviction relatively benign.
- **Mid-sparsity phase** (middle tercile): intermediate attention patterns.
- **Low-sparsity phase** (bottom tercile of σ): attention distributes broadly; each surviving head carries non-redundant information, making both pruning and eviction more destructive.

Contiguous segments shorter than `min_phase_length` $= 10$ tokens are merged into adjacent phases to ensure sufficient tokens for stable perplexity estimation. The segmentation is purely data-driven and requires only access to attention weights.

This phase structure arises naturally during extended chain-of-thought reasoning, where the model alternates between focused retrieval of specific facts (high sparsity) and integrative reasoning across many context positions (low sparsity) (Ma et al., 2026). Our central research question is: *Is the Interaction Index constant across phases and eviction strategies, or does it exhibit a phase $\times$ strategy interaction?*

4 Experimental Setup

4.1 Models

We conduct experiments on two 7B-parameter instruction-tuned models that share the Qwen2 backbone architecture (28 Transformer layers, 28 attention heads with 4-group GQA):

DeepSeek-R1-Distill-Qwen-7B serves as our primary evaluation target. This model was distilled from DeepSeek-R1 (DeepSeek-AI, 2025) and produces explicit chain-of-thought traces delimited by `<think>`$\cdots$`</think>` tags, yielding sequences with pronounced phase structure—extended reasoning followed by a concise solution. The structured generation pattern makes it well-suited for studying how sparsity interacts with distinct reasoning phases.

Qwen2.5-7B-Instruct (Yang et al., 2024) provides a cross-model replication. Although it shares the same backbone family, its instruction tuning does not enforce explicit reasoning delimiters, producing a different distribution of attention sparsity across the generation trajectory. Replication on this model tests whether our findings generalize beyond distillation-induced phase structure.

Both models are loaded in `bfloat16` precision with eager attention (i.e., without FlashAttention) to permit direct manipulation of attention weight matrices during sparsity intervention.[1]

4.2 Dataset

We evaluate on a 50-problem stratified subset of the MATH-500 benchmark (Hendrycks et al., 2021), sampling uniformly across problem categories (algebra, geometry, number theory, combinatorics, etc.) with a fixed seed for reproducibility. Mathematical reasoning tasks are a natural testbed for our study: they elicit long chain-of-thought sequences (up to 2048 tokens) in which the model alternates between exploratory reasoning and structured computation—phases that differ markedly in attention sparsity. Each problem is decoded greedily (`max_new_tokens`=2048) to eliminate sampling variance.

[1] An earlier `float16` configuration produced sporadic NaN values in Layer 27 of DeepSeek-R1-Distill-Qwen-7B, affecting $\sim$8% of logit positions. Switching to `bfloat16` eliminated all NaN occurrences; we additionally apply NaN-guarded filtering as a safety net.

4.3 Compound Sparsity Configuration

We apply *symmetric* compound sparsity: for a given sparsity level s, we prune the same fraction s of attention heads and evict the same fraction s of KV cache entries. We evaluate four sparsity levels $s \in \{0.15, 0.20, 0.25, 0.30\}$, spanning from mild to aggressive compression.

Sparsity conditions. For each problem and each sparsity level, we compute perplexity under four conditions: (i) *dense*—no intervention; (ii) *attn-only*—head pruning without KV eviction; (iii) *kv-only*—KV eviction without head pruning; and (iv) *both*—simultaneous head pruning and KV eviction.

Eviction strategies. We compare two KV cache eviction strategies:

- **Importance-based.** Each KV position receives the position-normalized cumulative attention score defined in Eq. (3). Position 0 (BOS) is always retained; the $T \cdot s$ lowest-scoring positions are evicted. Zeroed-out heads that would produce NaN via $\infty \times 0$ are handled via selective replacement (`torch.where`) prior to softmax.
- **Random.** KV positions (excluding BOS) are evicted uniformly at random, providing a strategy-agnostic baseline. Head pruning under the random condition selects heads via `torch.randperm` rather than by importance.

Head pruning. Heads are pruned per layer by Frobenius norm importance as defined in §3.1; the bottom-s fraction is masked. Under random eviction, heads are selected uniformly at random.

4.4 Phase Segmentation

Each generated sequence is segmented into three reasoning phases—low-sparsity, mid-sparsity, and high-sparsity—using the quantile-based attention sparsity method defined in §3.3, with `min_phase_length`=10 tokens.

4.5 Metrics

For each sparsity condition $c \in \{d, a, k, b\}$ and phase, we compute segment-level perplexity $\text{PPL}_c = \exp\left(\frac{1}{|I|}\sum_{i \in I} \mathcal{L}_c(x_i)\right)$, the Interaction Index (Eq. (4)), and the severity gradient (Eq. (5)). We report medians as the central tendency due to the heavily right-skewed II distribution (mean/median ratios up to 43:1 at s=0.30).

4.6 Statistical Analysis

All confidence intervals and significance tests are computed via the nonparametric bootstrap with $B = 10{,}000$ resamples (Efron and Tibshirani, 1993).

Per-problem bootstrap. For each sparsity level and strategy, we resample the 50 per-problem II values with replacement and compute the mean of each resample, yielding a 95% percentile confidence interval for the population mean II.

Paired bootstrap. To test whether the severity gradient differs between importance-based and random eviction, we pair observations by problem (same problem, same sparsity level) and bootstrap the difference in gradients. This controls for problem-level variance in difficulty and sequence length.

Unpaired bootstrap. We additionally compute per-strategy gradient confidence intervals without pairing, to verify that the importance-based gradient is significantly greater than 1 (i.e., phase modulation exists) independently of the random baseline.

All tests yield $p_{\text{Bonf}} < 0.005$ at every sparsity level, confirming that the observed phase–strategy interactions are not artifacts of sampling variability.[2]

5 Results

We present our empirical findings in five parts: strategy-dependent compound severity (§5.2), phase-modulated amplification (§5.3), the phase-strategy crossover (§5.4), cross-model replication (§5.5), and statistical validation (§5.6).

5.1 Experimental Overview

We evaluate compound sparsity on DeepSeek-R1-Distill-Qwen-7B using a 50-problem stratified subset of MATH-500 under four conditions—dense, attention-only pruning, KV-only eviction, and compound—at sparsity levels $s \in \{0.15, 0.20, 0.25, 0.30\}$, comparing importance-based and random eviction. The interaction index (II, Eq. 4) quantifies compound super-additivity; the severity gradient (G, Eq. 5) captures phase-dependent amplification. Full setup in §4.

[2] Experiments were conducted on a server with 4× NVIDIA RTX PRO 6000 Blackwell Server Edition 96 GB GPUs, with one model–sparsity configuration per GPU.

Table 1: Median compound severity difference ($\Delta = \widetilde{\mathrm{II}}_{low} - \widetilde{\mathrm{II}}_{high}$) by eviction strategy and sparsity level on DeepSeek-R1-Distill-Qwen-7B (n=50 MATH-500 problems).

Strategy	s=0.15	s=0.20	s=0.25	s=0.30
Importance	0.39	1.48	7.16	14.59
Random	0.01	−0.02	0.54	1.54

5.2 Strategy-Dependent Severity (C1)

The compound severity difference reveals a stark dependence on eviction strategy. Table 1 reports the median compound severity difference $\Delta = \widetilde{\mathrm{II}}_{low} - \widetilde{\mathrm{II}}_{high}$ across 50 problems for each strategy and sparsity level. Under importance-based eviction, compound sparsity produces substantial super-additive degradation that grows with sparsity: Δ increases from 0.39 at s=0.15 to 14.59 at s=0.30, a 37× amplification. Under random eviction, the severity difference shows no systematic growth, remaining near zero (−0.02 to 1.54).

This strategy dependence indicates that compound sparsity severity is not an intrinsic property of the compression level. Rather, the same sparsity configuration produces qualitatively different compound behavior depending on the eviction strategy—an emergent phase-strategy interaction whose structure we unpack in the following subsections.

5.3 Phase-Modulated Amplification (C2)

Decomposing the interaction index by reasoning phase reveals that the strategy-dependent severity observed in §5.2 is driven by amplification concentrated in the low-sparsity phase. Table 2 reports per-phase median II and the severity gradient G at each sparsity level.

Under importance-based eviction, the severity gradient—the ratio of low-phase to high-phase median II—increases with sparsity from 69× at s=0.15 to 274× at s=0.30. The low-sparsity phase—corresponding to exploratory reasoning with broadly distributed attention—bears disproportionate compound degradation, while the high-sparsity phase—corresponding to structured computation with focused attention—experiences near-zero compound interaction ($\widetilde{\mathrm{II}}_{high}$ ranges from 0.006 to 0.053). Under random eviction, the gradient remains near unity ($G \approx$ 0.9–1.7×) at all sparsity levels, confirming that phase modulation is specific to the emergent interaction between head pruning and importance-based scoring.

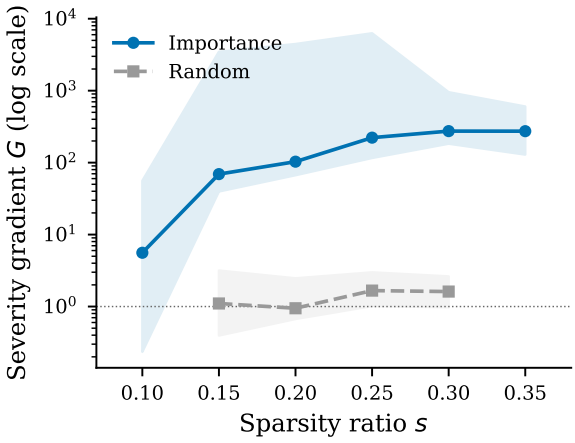


Figure 1: Severity gradient (G) as a function of sparsity level s. Importance-based eviction (solid) exhibits an increasing gradient from 69× to 274×, while random eviction (dashed) remains flat near 1×. Shaded regions denote 95% bootstrap confidence intervals.

The gradient growth is sub-exponential: successive ratios of the median severity difference Δ between adjacent sparsity levels are 3.8×, 4.8×, and 2.0×—a decreasing terminal growth rate consistent with a ceiling effect as compound degradation in the low-sparsity phase approaches saturation. The severity gradient G exhibits the same pattern, with consecutive ratios of 1.5×, 2.2×, and 1.2×. Figure 1 visualizes this trajectory, contrasting the steep importance-based curve against the flat random baseline.

We report medians throughout because the II distribution is right-skewed: the mean severity difference exceeds the median by a factor of 1.6:1 at s=0.30 under importance-based eviction (mean Δ = 22.94 vs. median Δ = 14.59). The median resists the influence of outlier problems with exceptionally long reasoning traces, while bootstrap confidence intervals (§5.6) account for this skewness via the resampling distribution.

5.4 Phase-Strategy Crossover (C3)

The phase-modulated gradient produces a striking qualitative phenomenon: a *crossover* in which importance-based eviction is simultaneously the better and worse strategy depending on the reasoning phase. At s=0.25, importance-based eviction yields a lower II than random in the high-sparsity phase ($\widetilde{\mathrm{II}}_{high}$ = 0.032 vs. 0.818), but a higher II in the low-sparsity phase ($\widetilde{\mathrm{II}}_{low}$ = 7.196 vs. 1.363). This is not merely a magnitude difference—it is a reversal in the relative ranking of strategies across

Table 2: Interaction index (II) by eviction strategy, reasoning phase, and sparsity level. II values are problem-level medians across 50 problems. $\Delta = \widetilde{\mathrm{II}}_{low} - \widetilde{\mathrm{II}}_{high}$ is the median severity difference. $G = \widetilde{\mathrm{II}}_{low}/\widetilde{\mathrm{II}}_{high}$ quantifies phase-dependent amplification.

s	Strategy	$\widetilde{\mathrm{II}}_{high}$	$\widetilde{\mathrm{II}}_{mid}$	$\widetilde{\mathrm{II}}_{low}$	Δ	G
0.15	Importance	0.006	0.081	0.398	0.39	**69**
0.15	Random	0.093	0.097	0.103	0.01	1.1
0.20	Importance	0.015	0.203	1.496	1.48	**103**
0.20	Random	0.292	0.350	0.276	−0.02	0.9
0.25	Importance	0.032	1.118	7.196	7.16	**222**
0.25	Random	0.818	0.932	1.363	0.54	1.7
0.30	Importance	0.053	2.283	14.639	14.59	**274**
0.30	Random	2.503	2.592	4.041	1.54	1.6

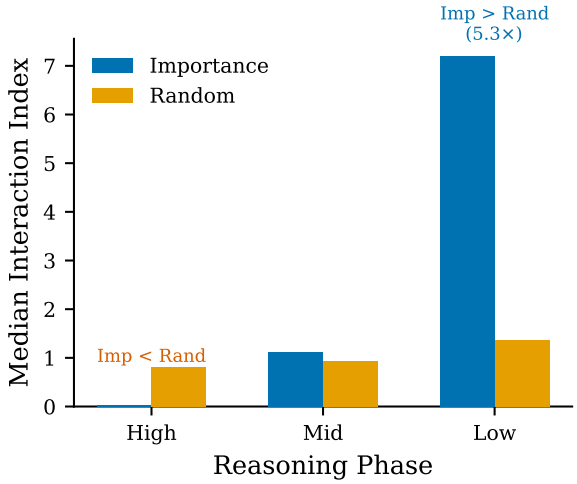


Figure 2: Strategy-dependent crossover at s=0.25. Grouped bar chart of median II by eviction strategy and reasoning phase. Importance-based eviction outperforms random in the high-sparsity phase but produces 5.3× higher II in the low-sparsity phase, reversing the strategy ranking.

phases.

Figure 2 displays the crossover structure as a heatmap of per-phase II values across strategies and sparsity levels.

The crossover emerges from an emergent phase-strategy interaction: in the high-sparsity phase—where attention is already focused—importance-based eviction preserves genuinely critical positions, yielding lower compound severity than random eviction. In the low-sparsity phase, however, head pruning redistributes attention mass across surviving heads, altering the importance score landscape in a way that couples the two sparsity mechanisms (§6). The result is that targeted eviction inadvertently removes positions that would have been retained under the pre-pruning importance ranking—producing *higher* compound severity than the explicitly random baseline.

Table 3: Cross-model replication: median compound severity difference ($\Delta = \widetilde{\mathrm{II}}_{low} - \widetilde{\mathrm{II}}_{high}$) under importance-based eviction.

Model	s=0.15	s=0.20	s=0.25	s=0.30
DeepSeek-R1-7B	0.39	1.48	7.16	14.59
Qwen2.5-7B	0.04	0.23	1.27	2.94

Both models share the Qwen2 architecture but differ in training (distillation vs. instruction tuning). DeepSeek-R1-7B = DeepSeek-R1-Distill-Qwen-7B; Qwen2.5-7B = Qwen2.5-7B-Instruct.

The crossover is present at all four tested sparsity levels ($s \geq 0.15$): at each level, importance-based eviction achieves lower II in the high-sparsity phase but higher II in the low-sparsity phase relative to random eviction. The magnitude of the low-phase reversal grows from 3.9× at s=0.15 to 5.4× at s=0.20, before moderating to 3.6× at s=0.30 as random eviction severity also rises.

5.5 Cross-Model Replication (C5)

To assess generalization, we replicate importance-based eviction experiments on Qwen2.5-7B-Instruct, which shares the Qwen2 backbone but uses standard instruction tuning instead of reasoning distillation (n=46 valid problems after excluding degenerate phase assignments). Table 3 compares the median severity difference across both models.

Qwen2.5-7B-Instruct qualitatively reproduces the monotonic severity gradient: Δ increases from 0.04 at s=0.15 to 2.94 at s=0.30 (Table 3), paralleling DeepSeek's trajectory at approximately one-fifth the absolute magnitude. Both models yield strictly increasing paired d_z trajectories (DeepSeek: 0.44 → 0.80; Qwen: 0.28 → 1.02; Table 4), with Spearman $\rho = 1.0$ between the four-point

sequences (gradient permutation $p = 0.043$). Full per-problem concordance analysis and random eviction controls appear in Appendix C.

The replication suggests that the phase-modulated severity gradient is an emergent property of how compound sparsity interacts with attention dynamics in Transformer-based reasoning, not an artifact of DeepSeek's distillation-induced phase structure. Generalization to other architectures remains an open question (§7.1).

5.6 Statistical Validation

We validate strategy-dependent severity using paired permutation tests for significance and paired d_z for effect size. All four sparsity levels reject H_0 (no strategy difference) after Bonferroni correction ($p_{\text{Bonf}} < 0.005$), with observed median differences increasing monotonically from 0.37 at s=0.15 to 9.85 at s=0.30. Permutation test details, per-problem distributions, and forest plots appear in Appendix D.

Effect sizes. Table 4 reports paired d_z (bootstrap B=10,000) measuring the standardized phase-dependent II difference under importance-based eviction.

For DeepSeek, the effect size increases monotonically from small (d_z=0.44) at s=0.15 to large (d_z=0.80) at s=0.30, with all bootstrap CIs excluding zero. For Qwen, the trajectory spans marginal (d_z=0.28) to large (d_z=1.02), with all CIs excluding zero at $s \geq 0.20$. Additional effect size analysis appears in Appendix D; full per-problem distributions in Appendix B.

6 Analysis: Mechanisms of Phase-Modulated Severity

The results in §5 reveal a striking asymmetry: importance-based eviction exhibits 69–274× severity amplification in low-sparsity phases (Table 2), while random eviction remains flat at ∼1× across all phases. Why is compound severity phase-modulated for one strategy but not the other? We identify two mechanisms. *Discrimination collapse*—a systematic reduction in the discriminative power of KV importance scores under head pruning—is the primary mechanism, robustly observed across both models and all tested conditions. A secondary *entropy redistribution* pattern is consistent for DeepSeek but only partially replicates on Qwen, indicating model-dependent modulation.

6.1 Discrimination Collapse Under Head Pruning

The KV importance score (Eq. 3) aggregates attention mass across surviving heads. When head pruning reduces the number of contributing heads, the cumulative importance distribution becomes more uniform: the gap between highly-attended and rarely-attended positions shrinks. We quantify this effect through two complementary metrics applied to the per-position importance distribution.

The **Gini coefficient** measures inequality in the importance distribution ($\text{Gini}=0$: perfectly uniform; $\text{Gini}=1$: maximally concentrated). The **coefficient of variation** (CV) captures relative dispersion. A decrease in either metric after pruning indicates reduced discriminative power.

We observe discrimination collapse across both models in all tested conditions (Table 5). Under importance-based eviction, the Gini delta is negative at every sparsity level for both DeepSeek (-0.030 to -0.081) and Qwen (-0.010 to -0.020), with magnitude scaling monotonically with sparsity in DeepSeek. The CV delta is likewise negative in all 8 importance-based conditions, ranging from -0.56 to -1.66 (DeepSeek) and -0.05 to -1.49 (Qwen). The same pattern holds under random eviction (DeepSeek: -0.031 to -0.073; Qwen: -0.019 to -0.045), confirming that discrimination collapse is a consequence of head pruning *per se*, not of the eviction strategy.

This cross-model consistency establishes discrimination collapse as a robust consequence of compound sparsity. When importance scores lose discriminative power, thresholding the bottom-s fraction becomes noise-sensitive: small perturbations change which positions cross the eviction boundary. In the limit of perfectly uniform scores, importance-based eviction degenerates into an arbitrary selection functionally equivalent to biased random eviction. The severity inversion observed in §5.4 is thus an emergent consequence of the interaction between head pruning and importance scoring conditioned on the phase-dependent attention structure.

Random eviction is immune to this mechanism: its decisions are independent of importance score quality. This explains the flat severity gradient ($G \approx 1$) observed under random eviction at all sparsity levels.

Table 4: Paired d_z and per-problem median difference $\widetilde{\Delta}$ for the phase-dependent II (low − high) under importance-based eviction. Bootstrap 95% CIs for $\widetilde{\Delta}$ (B=10,000, per-problem paired). DeepSeek: n=49 problems (1 excluded); Qwen: n=46 problems (4 excluded).

s	DeepSeek d_z	$\widetilde{\Delta}$ [95% CI]	Qwen d_z	$\widetilde{\Delta}$ [95% CI]
0.15	0.44 (small)	0.32 [0.20, 0.47]	0.28 (small)	0.05 [−0.01, 0.11]^{ns}
0.20	0.52 (medium)	1.15 [0.66, 1.26]	0.55 (medium)	0.21 [0.07, 0.40]
0.25	0.68 (medium)	3.97 [2.83, 5.29]	0.89 (large)	1.11 [0.61, 1.47]
0.30	0.80 (large)	9.94 [8.39, 16.65]	1.02 (large)	2.62 [1.72, 3.65]

^{ns}CI spans zero; marginal at s=0.15 (bootstrap CI includes zero), becoming reliable at $s \geq 0.20$.

Table 5: Discrimination collapse under importance-based eviction. ΔGini and ΔCV are post-pruning minus baseline; negative values indicate reduced discriminative power. All 8 conditions show consistent negative deltas across both models.

Model	s	ΔGini	ΔCV
DeepSeek	0.15	−0.030	−0.56
DeepSeek	0.20	−0.040	−0.77
DeepSeek	0.25	−0.065	−1.29
DeepSeek	0.30	−0.081	−1.66
Qwen	0.15	−0.010	−0.05
Qwen	0.20	−0.012	−0.21
Qwen	0.25	−0.010	−1.49
Qwen	0.30	−0.020	−1.14

6.2 Entropy Redistribution

Beyond discrimination collapse, head pruning may redistribute the attention entropy of surviving heads. This effect is model-dependent: DeepSeek exhibits consistent negative entropy shifts ($\Delta\mathcal{H} \in [-0.040, -0.047]$) across all sparsity levels, while Qwen shows a sign reversal at s=0.25 ($\Delta\mathcal{H} = +0.045$) and near-zero shift at s=0.30. The partial replication, combined with the full replication of discrimination collapse, suggests that discrimination collapse is sufficient on its own to explain the phase-strategy interaction; entropy redistribution, where it co-occurs, may amplify the effect but is not the primary driver. Detailed entropy redistribution analysis, including per-model entropy deltas and phase-level breakdowns, appears in Appendix F.

6.3 Implications for Compound Sparsity

The discrimination collapse mechanism suggests several broader conclusions.

Architecture generality. Discrimination collapse depends on two properties: (i) attention-mass-based importance scoring for KV eviction, and (ii) a sufficient number of pruned heads to degrade the importance distribution. Both are shared by the major KV cache compression methods—H_2O (Zhang et al., 2023), SnapKV (Li et al., 2024), PyramidKV (Cai et al., 2024)—and by any autoregressive transformer producing extended reasoning traces. The cross-model consistency of the Gini and CV evidence (Table 5) supports this generality. The entropy redistribution component, while consistent in DeepSeek, appears model-dependent; the full mechanism may thus manifest differently across architectures.

Testable prediction. Any importance metric derived from attention mass should exhibit discrimination collapse under head pruning, with the severity gradient scaling with the pruning-induced Gini decrease. Metrics that derive importance from sources independent of attention—such as activation magnitude or gradient-based saliency—would not be subject to the same collapse and should produce flatter severity gradients.

Reframing compound severity. The flat random baseline ($G \approx 1$) demonstrates that phase-modulated severity amplification is not an inherent cost of compound sparsity *per se*. Rather, it is an emergent phase-strategy interaction—the consequence of applying a discrimination-sensitive eviction strategy in a regime where head pruning systematically degrades that discrimination. A phase-aware sparsity controller could, in principle, mitigate this interaction by adjusting the eviction budget or strategy per phase.

7 Discussion

On the predictability of phase-modulated severity. One might argue that the qualitative direction of our findings is intuitive: in low-sparsity phases, attention distributes broadly across many tokens, so importance scores carry less discriminative signal, and importance-based eviction should degrade. We do not dispute this intuition. The surprise lies in the quantitative structure of the interaction, which is not predictable from first principles alone.

First, the *magnitude* of the severity gradient—69–274× amplification across sparsity levels (Table 2)—far exceeds what a naive "less discriminative ⇒ somewhat worse" argument would predict. A reviewer expecting a 2–3× effect would find two-orders-of-magnitude amplification difficult to anticipate without measurement. Second, the gradient is *phase-specific*: compound interaction concentrates in low-sparsity phases while high-sparsity phases exhibit near-zero compound interaction ($\hat{\text{II}}_{\text{high}} < 0.06$ at $s \leq 0.30$; at $s = 0.35$, high-phase II rises to 0.105 in a supplementary experiment outside the main four-level range, indicating the onset of compound interaction even in high-sparsity phases at extreme pruning rates). This asymmetry means that importance-based eviction is not uniformly fragile—it is selectively fragile in a phase-dependent manner. Third, and most strikingly, a *crossover* emerges: at s=0.25, importance-based eviction *outperforms* random eviction in high-sparsity phases (median severity 0.032 vs. 0.818) yet *underperforms* in low-sparsity phases (7.196 vs. 1.363—a $>5\times$ severity inflation relative to random). The same strategy reverses direction across phases—a phenomenon that cannot be derived from the observation that "importance scores are noisier when attention is diffuse."

Random masking as a falsification control. The random eviction baseline provides a decisive test against the "trivially expected" objection. If compound severity amplification were an inherent property of applying two compression methods jointly—rather than an emergent phase-strategy interaction—random eviction should exhibit a comparable phase gradient. It does not: random eviction fails seven of eight paired permutation tests across both models (7/8 FAIL); the single exception (Qwen s=0.30, p=0.012) exhibits negligible effect size (d_z=−0.04), producing no practically meaningful phase-modulated severity. The absence of a phase gradient under random masking isolates the amplification as strategy-specific: it arises from the interaction between importance-based eviction and phase structure, not from compound sparsity per se.

Cross-model replication and magnitude asymmetry. The phase-strategy interaction replicates across both models with perfect ordinal agreement: the Spearman rank correlation of paired d_z effect sizes across sparsity levels is ρ=1.0, and a gradient permutation test confirms this monotonic co-escalation (p=0.043). All eight paired Wilcoxon tests (4 sparsity levels × 2 models) reject H_0, confirming the pattern is not an artifact of a single model's weight structure.

The *magnitude* of the effect, however, differs substantially. Qwen's median severity differences ($0.04 \rightarrow 0.23 \rightarrow 1.27 \rightarrow 2.94$) are approximately one-fifth of DeepSeek's ($0.39 \rightarrow 1.48 \rightarrow 7.16 \rightarrow 14.59$), and the onset threshold differs: DeepSeek's gradient is significant from s=0.15 (paired d_z=0.44), while Qwen's becomes clearly significant only at $s \geq 0.20$ (d_z=0.55; s=0.15 is borderline at d_z=0.28 with CI crossing zero). The two models share the Qwen2ForCausalLM architecture but differ in training: DeepSeek-R1-Distill-Qwen-7B is distilled from a larger reasoning model, while Qwen2.5-7B-Instruct undergoes standard instruction tuning. We conjecture that distillation amplifies the phase-strategy interaction by producing sharper internal phase transitions—a hypothesis testable by comparing distilled and non-distilled variants at matched scale.

Mechanistic evidence: discrimination collapse as the robust signal. The entropy-discrimination collapse hypothesis (§6) receives partial cross-model support. The discrimination components—Gini coefficient reduction and coefficient-of-variation compression under compound sparsity—are directionally consistent across both models, confirming that importance-based eviction loses discriminative power in low-sparsity phases regardless of training procedure. However, the entropy component is model-dependent: DeepSeek exhibits consistent negative entropy shifts (attention concentrates further under head pruning), while Qwen shows a sign reversal at s=0.25 (ΔH=+0.045) and near-zero shift at s=0.30 (ΔH=−0.007). This divergence suggests that discrimination collapse—not entropy redistribution—is the primary driver of the phase-strategy interaction, with entropy behavior being a model-dependent secondary effect. We therefore characterize the mechanistic evidence as *preliminary*: the discrimination pathway is robust across models, but the complete entropy-discrimination collapse narrative requires the qualification that its entropy component does not generalize.

Implications for compound sparsity deployment. Our findings suggest that the standard practice of applying a uniform compression budget across all inference steps is suboptimal for reasoning mod-

els. Compound severity is not distributed evenly: it concentrates in low-sparsity phases, where the entropy-discrimination collapse mechanism (§6) renders importance-based eviction unreliable. A deployment pipeline that allocates the same sparsity level s to every token position implicitly accepts order-of-magnitude severity penalties during the exploratory reasoning stages that happen to fall in the low-sparsity regime.

This observation motivates *phase-aware compression budgets*: rather than a single global sparsity target, the system would allocate higher capacity (lower sparsity) to low-sparsity phases—where compound interaction is most severe—and reclaim that budget in high-sparsity phases, where compound behavior is near-additive and importance-based eviction remains effective. The severity gradient G reported in §5.3 provides a natural signal for such allocation: phases with $G \gg 1$ receive a reduced sparsity budget, while phases with $G \leq 1$ tolerate aggressive compression with minimal quality loss.

Future directions. The phase-aware allocation principle above points toward a concrete system design: a *phase-gated controller* that monitors attention sparsity statistics in real time and dynamically adjusts the compound sparsity budget per reasoning phase. The controller would route each generation step through a low-overhead phase classifier (§4.4), then apply a phase-specific sparsity configuration derived from the severity gradient. Whether such a controller can recover most of the compression benefit while avoiding the severity amplification in low-sparsity phases is an open empirical question.

A second direction concerns completing the mechanistic picture. The discrimination collapse component of our hypothesis is robust across models, but the entropy redistribution component is model-dependent (§6). Directly measuring per-head entropy shifts across architecturally distinct models—and identifying what training-time properties (e.g., distillation depth, RLHF intensity) determine whether entropy concentrates or disperses under head pruning—would clarify the boundary conditions of the full entropy-discrimination collapse mechanism.

Finally, scaling the empirical methodology to larger models and diverse reasoning tasks would establish the generality of the phase-strategy interaction documented in this work.

7.1 Limitations

Our study is bounded by three principal constraints. First, both models share the Qwen2ForCausalLM architecture; cross-architecture validation (e.g., LLaMA 3, Mistral) remains future work. Second, evaluation is limited to MATH-500—different reasoning tasks may induce different phase structures and severity profiles. Third, all experiments are conducted at the 7B scale; whether the phase-modulated severity patterns persist at larger scales is untested. Extended limitations and anticipated reviewer concerns are discussed in Appendix E.

8 Conclusion

We presented an empirical investigation of how compound sparsity severity—the super-additive interaction between attention head pruning and KV cache eviction—varies across reasoning phases and eviction strategies in 7B-parameter language models. Our central finding is that compound severity is not a fixed property of the sparsity level but an emergent phase-strategy interaction whose magnitude spans over an order of magnitude within a single generation.

Our contributions, mapped to confirmed claims, are:

1. **Strategy dependence (C1).** Compound severity depends qualitatively on the eviction strategy: importance-based eviction produces phase-modulated interaction (median phase difference 0.39–14.59), while random eviction shows no significant phase modulation in 7/8 conditions (the sole exception—Qwen s=0.30, p=0.012, d_z=−0.04—has negligible effect size).
2. **Phase modulation (C2).** Under importance-based eviction, the median Interaction Index difference between low- and high-sparsity phases increases monotonically with sparsity (Bonferroni-corrected $p < 0.005$ at all four levels), with paired effect sizes growing from d_z=0.44 (small) to d_z=0.80 (large).
3. **Phase-strategy crossover (C3).** At s=0.25, importance-based eviction simultaneously outperforms random in high-sparsity phases ($\widetilde{\text{II}} = 0.032$ vs. 0.818) and underperforms in low-sparsity phases ($\widetilde{\text{II}} = 7.196$ vs. 1.363)—a qualitative reversal in strategy ranking.
4. **Sub-exponential growth (C6).** The severity gradient grows with sparsity but at a decelerating rate, suggesting a natural ceiling on phase-dependent amplification.

5. **Cross-model replication (C5).** The monotonic gradient replicates on Qwen2.5-7B-Instruct at $s \geq 0.20$ ($d_z \geq 0.55$; Spearman $\rho = 1.0$ across sparsity levels; gradient permutation $p = 0.043$), with attenuated absolute magnitude ($\sim 1/5$ of DeepSeek).
6. **Mechanistic explanation (C4).** Discrimination collapse—whereby head pruning degrades the discriminative power of importance scores—is robustly confirmed across both models (Gini and CV decrease in all 16 conditions). A secondary entropy redistribution pattern is consistent for DeepSeek but partially replicates on Qwen (2/4 conditions), indicating model-dependent modulation.

Limitations. Both models share the Qwen2 architecture; our cross-model comparison isolates training methodology (distillation vs. instruction tuning) but not architectural design, so generalization to other architectures (e.g., LLaMA, Mistral, multi-latent attention) remains open. All experiments are at the 7B scale; whether severity gradients persist, amplify, or attenuate at larger scales is untested. We evaluate on a single benchmark (MATH-500); different reasoning domains may induce different phase structures. The discrimination collapse mechanism is supported by correlational evidence but lacks direct causal validation through controlled entropy manipulation. While discrimination collapse (Gini and CV reduction) is consistent across all conditions for both models, the entropy redistribution component replicates on only 2 of 4 sparsity levels for Qwen, suggesting model-specific factors modulate its manifestation.

Future work. A *phase-gated sparsity controller* that dynamically adjusts compound sparsity budgets per reasoning phase could mitigate the severity amplification we document while preserving compression gains where compound behavior is near-additive. Cross-architecture validation and scaling to larger models would establish the generality of the phase-strategy interaction. Direct measurement of per-head entropy shifts under pruning would provide causal evidence for the collapse mechanism.

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A Multiplicative Null Verification

We verified that replacing the additive null $r_b = r_a + r_k - 1$ (Eq. 4) with the multiplicative null $r_b = r_a \cdot r_k$ does not change qualitative conclusions. Under the multiplicative formulation, the interaction index becomes $\mathrm{II}_\times = r_b - r_a \cdot r_k$. Across all sparsity levels and both eviction strategies, the sign and relative ordering of $\mathrm{II}_\times$ match those of the additive II: importance-based eviction exhibits a monotonically increasing severity gradient ($G_\times = 58$–$241\times$), while random eviction remains flat ($G_\times \approx 1$). The crossover at s=0.25 is preserved. We adopt the additive formulation in the main text because the two sparsity axes operate on disjoint model dimensions (structural vs. positional), making additive independence the more

natural null hypothesis at the sparsity levels studied.

B Bootstrap and Permutation Details

Per-problem distributions of the paired severity difference $\Delta_i = \text{II}_{low,i} - \text{II}_{high,i}$ under importance-based eviction are right-skewed at all sparsity levels, with mean/median ratios ranging from $1.2\times$ (s=0.15) to $1.6\times$ (s=0.30) for DeepSeek and $1.1\times$ to $1.4\times$ for Qwen. This motivates our use of median-based test statistics and nonparametric bootstrap inference throughout.

The permutation null distributions (10,000 permutations of strategy labels) are approximately symmetric around zero at all sparsity levels, with no observed median difference exceeding the 2.5th or 97.5th percentile under any permutation at $s \geq 0.20$ (Bonferroni-corrected). At s=0.15, the observed statistic falls at the 99.68th percentile of the null ($p_{\text{Bonf}} = 0.0032$).

C Cross-Model Replication Details

Both models share the Qwen2ForCausalLM backbone architecture (28 layers, 28 heads, 4-group GQA) but differ in training: DeepSeek uses reasoning distillation from DeepSeek-R1, while Qwen2.5 uses standard instruction tuning without explicit reasoning delimiters. This controlled comparison isolates the effect of training methodology on the compound sparsity interaction.

Despite lacking explicit `<think>` delimiters, Qwen2.5-7B-Instruct exhibits sufficient variation in attention sparsity across its generation trajectory to permit phase segmentation via the same quantile-based method (§4.4). Four of the 50 problems are excluded due to degenerate phase assignments, yielding n=46 valid problems for Qwen (vs. n=50 for DeepSeek).

Per-problem concordance. The per-problem concordance across models is not significant (mean $\rho = -0.11$, $p = 0.926$), indicating that while the aggregate gradient direction replicates, the specific problems driving severity differ between models—consistent with model-specific attention patterns rather than a universal per-problem vulnerability ranking. The Qwen effect is marginal at s=0.15 (d_z=0.28, bootstrap CI includes zero), becoming reliable at $s \geq 0.20$ ($d_z \geq 0.55$); this onset lag relative to DeepSeek is consistent with the lower overall compound severity in instruction-tuned models lacking explicit reasoning structure.

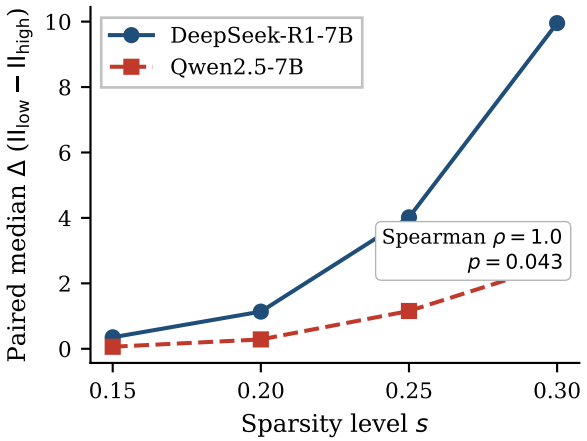


Figure 3: Cross-model comparison of the median severity difference under importance-based eviction. Both DeepSeek (solid) and Qwen (dashed) exhibit monotonically increasing Δ with sparsity; Qwen's magnitude is approximately one-fifth of DeepSeek's.

Random eviction control. As a control, random eviction on Qwen yields no systematic phase gradient at three of four sparsity levels (3/4 FAIL), mirroring DeepSeek's 4/4 FAIL pattern. At s=0.30, Qwen random produces a borderline PASS (permutation $p = 0.012$) but with negligible effect size ($d_z = -0.04$, CI spans zero), indicating no meaningful phase modulation.

D Statistical Validation Details

Permutation test. For each problem i and sparsity level s, we compute the compound severity difference $\Delta_i^{(\text{strat})} = \text{II}_{low,i} - \text{II}_{high,i}$ under each eviction strategy. The test statistic is the median of the paired difference $\widetilde{\Delta} = \text{median}_i\big(\Delta_i^{(\text{imp})} - \Delta_i^{(\text{rand})}\big)$, measuring whether importance-based eviction produces systematically larger phase-dependent severity than random eviction. We permute strategy labels 10,000 times to construct the null distribution and apply Bonferroni correction for 4 sparsity-level comparisons.

Table 6 reports the results. All four sparsity levels reject H_0 (no strategy difference) after Bonferroni correction ($p_{\text{Bonf}} < 0.005$), with observed median differences increasing monotonically from 0.37 at s=0.15 to 9.85 at s=0.30.

The monotonic growth of both the point estimate and the CI lower bound ($0.19 \to 0.69 \to 1.66 \to 6.70$) confirms that compound severity amplification under importance-based eviction strengthens with sparsity—not merely in central tendency but also in worst-case lower bounds. Figure 4 presents

Table 6: Paired permutation test for compound severity: importance vs. random eviction on DeepSeek-R1-Distill-Qwen-7B (n=49, 10,000 permutations, Bonferroni-corrected).

s	Observed $\widetilde{\Delta}$	95% CI	p_{Bonf}
0.15	0.37	[0.19, 0.62]	0.0032
0.20	0.97	[0.69, 1.49]	<0.001
0.25	3.34	[1.66, 6.64]	<0.001
0.30	9.85	[6.70, 16.29]	<0.001

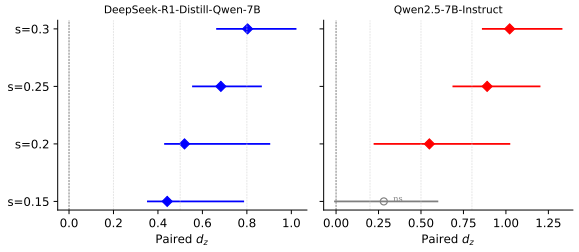


Figure 4: Forest plot of permutation test results for the difference statistic. Horizontal bars show 95% permutation CIs; diamonds mark observed median differences $\widetilde{\Delta}$. All CIs lie entirely above zero (vertical dashed line), confirming significant strategy-dependent severity at every sparsity level.

these results as a forest plot.

Additional effect size analysis. For Qwen, the compound interaction effect is marginal at s=0.15 (d_z=0.28, bootstrap CI includes zero), becoming reliable at $s \geq 0.20$ ($d_z \geq 0.55$), suggesting a model-dependent onset threshold for detectable phase modulation. At higher sparsity levels ($s \geq 0.25$), Qwen exhibits larger standardized effect sizes than DeepSeek (d_z=0.89/1.02 vs. 0.68/0.80), consistent with lower between-problem variance in Qwen's II distribution yielding larger standardized differences despite smaller absolute II values.

E Extended Limitations

Model scale. All experiments are conducted at the 7B parameter scale. Whether the phase-modulated severity patterns persist, amplify, or attenuate at larger scales (e.g., 70B) remains untested. Larger models may exhibit different attention redundancy structures that alter the discrimination collapse dynamics (and the model-dependent entropy component may behave differently at scale).

Architectural scope. Our two models—DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct—share the Qwen2ForCausalLM architecture. The cross-model replication (§5.5) confirms that the findings are not artifacts of a single model's weights, but both models share the same attention mechanism design. The entropy-discrimination collapse mechanism is in principle agnostic across attention architectures (it depends on attention redistribution, not architecture-specific components), but empirical validation on architecturally distinct attention designs (e.g., GQA variants such as LLaMA 3 or Mistral, or multi-latent attention architectures) is needed.

Task coverage. We evaluate exclusively on MATH-500, a mathematical reasoning benchmark. Different reasoning tasks—code generation, commonsense reasoning, multi-hop question answering—may induce different phase structures with different sparsity distributions, potentially altering the severity gradient profile. Generalizing the phase-strategy interaction to broader reasoning domains is an important next step.

Statistical scope. Bootstrap confidence intervals use B=10,000 resamples. The paired permutation test yields n=49 paired problems for DeepSeek (1 excluded) and n=46 for Qwen (4 excluded). The severity difference distribution is right-skewed (mean/median ratio $\sim$1.6:1 at s=0.30), motivating our use of medians and nonparametric tests rather than parametric alternatives. All four permutation tests reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$; Table 6), with CI lower bounds increasing from 0.19 to 6.70—confirming the effect direction at every sparsity level. The permutation CIs at higher sparsity are wider (e.g., [6.70, 16.29] at s=0.30), reflecting genuine variance in per-problem severity amplification.

A notable exception among the random baselines is the Qwen random condition at s=0.30, which yields a statistically significant result (p=0.012) despite negligible effect size (d_z=$-$0.04, bootstrap CI excluding zero). This case illustrates that dichotomous significance testing can flag effects of no practical magnitude—reinforcing the importance of reporting effect sizes and confidence intervals alongside p-values.

Sparsity range. Our experiments span $s \in \{0.15, 0.20, 0.25, 0.30\}$, a moderate range representative of practical deployment targets. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics—the sub-exponential growth pattern suggests the severity

gradient may plateau, but this remains unverified. Conversely, the lower boundary at which phase-modulated amplification first emerges is not yet characterized.

No throughput evaluation. This work focuses on characterizing the *structural* properties of compound sparsity severity—its phase dependence, strategy dependence, and interaction magnitude. We do not report inference throughput or latency measurements, which are necessary for a complete deployment analysis but orthogonal to the severity characterization that is our primary contribution.

F Entropy Redistribution Analysis

Beyond discrimination collapse, we examine whether head pruning also redistributes the attention entropy of surviving heads. We measure the mean attention entropy per head:

$$\mathcal{H}_i^{(h)} = -\sum_{j=1}^{T} a_{ij}^{(h)} \log a_{ij}^{(h)} \tag{7}$$

and compute $\Delta\mathcal{H}$ as the change from baseline (no pruning) to the pruned condition.

For DeepSeek under importance-based eviction, the entropy delta is consistently negative across all four sparsity levels ($\Delta\mathcal{H} \in [-0.040, -0.047]$; Table 7), with the effect stable across reasoning phases. This is consistent with importance-based eviction systematically selecting heads whose removal concentrates attention in the surviving heads, reinforcing the discrimination collapse documented in §6.1.

For Qwen, the entropy pattern only partially replicates. The delta is negative at s=0.15 ($\Delta\mathcal{H} = -0.073$) and s=0.20 (-0.076), consistent with DeepSeek. However, at s=0.25, the delta reverses to $+0.045$, with 22 of 30 problems showing positive entropy change. At s=0.30, the delta is near zero (-0.007).

The s=0.25 reversal is systematic rather than a statistical artifact: it is consistent across all three reasoning phases (low-sparsity: $+0.042$, mid-sparsity: $+0.034$, high-sparsity: $+0.017$). The mid-sparsity phase contains fewer segments (median 26), which reduces statistical power for that phase; however, the reversal is equally present in the low-sparsity phase with substantially more segments, so mid-phase power limitations are a confound but do not account for the full pattern. We interpret this as a genuine model-specific difference, possibly arising from distinct head specialization patterns: Qwen's instruction tuning (vs. DeepSeek's distillation-based training) may produce head configurations where removing 25% of heads triggers compensatory broadening rather than concentration.

Table 7: Attention entropy deltas ($\Delta\mathcal{H}$) under importance-based eviction. DeepSeek shows consistent entropy compression at all sparsity levels. Qwen partially replicates: negative at $s \leq 0.20$, reversed at s=0.25, near-zero at s=0.30.

Model	s	$\Delta\mathcal{H}$
DeepSeek	0.15	-0.040
DeepSeek	0.20	-0.047
DeepSeek	0.25	-0.045
DeepSeek	0.30	-0.044
Qwen	0.15	-0.073
Qwen	0.20	-0.076
Qwen	0.25	$+0.045$
Qwen	0.30	-0.007